DTAG

MANUAL

Web Workbench · Technical User Guide

Survey-grounded digital respondents

General survey interaction, GSS ideology geometry, directional questioning and replayable experiments

Browser deployment | User and research-operator guide

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Status: documentation draft; deployed-build verification pending

Application URL https://dtag-workbench-gp7m2nymma-ue.a.run.app

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Scope

This manual covers only the password-protected, browser-accessed DTAG Workbench. It excludes local installation, command-line tools, APIs for direct operation, the separate PsiSim application and the DTAG Prolific human-participant application.

Verification boundary: the browser deployment could not be directly inspected from this authoring environment. All described controls are grounded in the connected DTAG web-interface source and its documented workflows. Before distribution as a production release, compare "About" → version/commit with the deployment and confirm the indicated controls on the live service.

0 How to read this manual

A practical operator guide and a technical interpretation reference, organized from ordinary interaction to GSS-only research tools.

Part	What it covers
Sections 1–3	Access, interface, survey choice, respondent creation and model conditioning
Sections 4–6	Question answering, evidence, decision landscape and two-respondent comparison
Sections 7–9	GSS ideology mathematics, directional questions and automatic push
Sections 10–12	Sequences, experimental reruns, statistics and exports
Sections 13–16	Settings, worked protocols, troubleshooting and interpretation
Appendices	Glossary, output dictionary, implementation and verification notes

Conventions

- UI names such as “Start respondent”, “Model evidence”, “Auto-push” and “Rerun ×N” are reproduced from the documented browser controls.
- Survey-dependent functions are explicitly distinguished: landscape and normal Q&A exist for all families; a directional ideology index and associated optimizer are GSS-only.
- The words “left” and “right” refer to the model’s designated GSS pole states, not a party label or a determination of a real individual’s affiliation.
- “State change” means a changed conditioning record inside the model; it does not mean persuasion or an observed human change.

At-a-glance capability matrix

Capability	GSS	WVS7	Afrobarometer	Eurobarometer
Survey-grounded chat	Yes	Yes	Yes	Yes
Evidence and distributions	Yes	Yes	Yes	Yes
Decision landscape	Yes	Yes	Yes	Yes
Two respondents	Yes	Yes	Yes	Yes
Ordered question sequence	Yes	Yes	Yes	Yes
Experiment replay / exports	Yes	Yes	Yes	Yes
Wave-specific ideology index	Yes	No	No	No
Directional next question and auto-push	Yes	No	No	No

Documented model inventory: 252 native survey models (35 GSS waves, 1 pooled WVS7, 9 Afrobarometer rounds and 207 Eurobarometer studies), in model release v0.2.1. Availability can depend on the deployed catalog and model-readiness state.

1 Purpose, architecture and operating boundary

DTAG is a conditional survey-model interface, not a free-form respondent simulator.

A DTAG digital respondent is a stateful collection of survey-item/answer assignments combined with a native Large Science Model (LSM) for a chosen survey. The LSM produces conditional answer distributions. A language model interprets the persona, maps natural-language questions to survey items and phrases outputs using the selected native survey-response anchors. The language model is not the authority for survey answer probabilities.

1.1 The interaction loop

1. Choose a supported survey model and describe the respondent (who, where, when).
2. Initialize valid survey-item/answer assignments to form the starting state.

3. Ask a natural-language question; map it to one or more native survey items.
4. Compute each item's conditional response distribution given the current state; sample or select an anchor.
5. Optionally update state, recompute dependent predictions, and (for GSS) recompute the ideology index.
6. Generate a readable answer linked to the native evidence; record the action for export and replay.

1.2 Scientific scope

Appropriate uses. Investigate model-conditional survey answers; compare modeled personas, waves and country contexts; inspect question-to-variable grounding; study how observed/assigned answers alter conditional predictions; explore adaptive question selection and its simulated trajectories.

Not supported. DTAG does not identify a real respondent, infer an individual's actual party membership, estimate a causal treatment effect, or prove a persuasive intervention changes beliefs. Its outputs inherit biases, coding choices and coverage limits of the training surveys and fitted models.

2 Access and interface orientation

All work described here takes place at the HTTPS Workbench URL; no local installation is required.

2.1 Sign in and verify the application

1. Open the DTAG Workbench URL in a current desktop browser.
2. Authenticate using the password supplied by the research administrator. The service uses a shared-password research-pilot login rather than separate user accounts.
3. Open "About" or click the DTAG mark in the top bar and record the reported application version/commit for reproducibility.
4. Check whether model download/loading is in progress. Start the respondent once the chosen model is ready; starting may also wait for model loading.

Active sessions and native model objects are process-local in the described Cloud Run deployment; an instance replacement or application restart can terminate a session. Export data before resetting, leaving an extended session or completing an experiment. Browser login and server-side respondent state are separate things.

2.2 Main screen

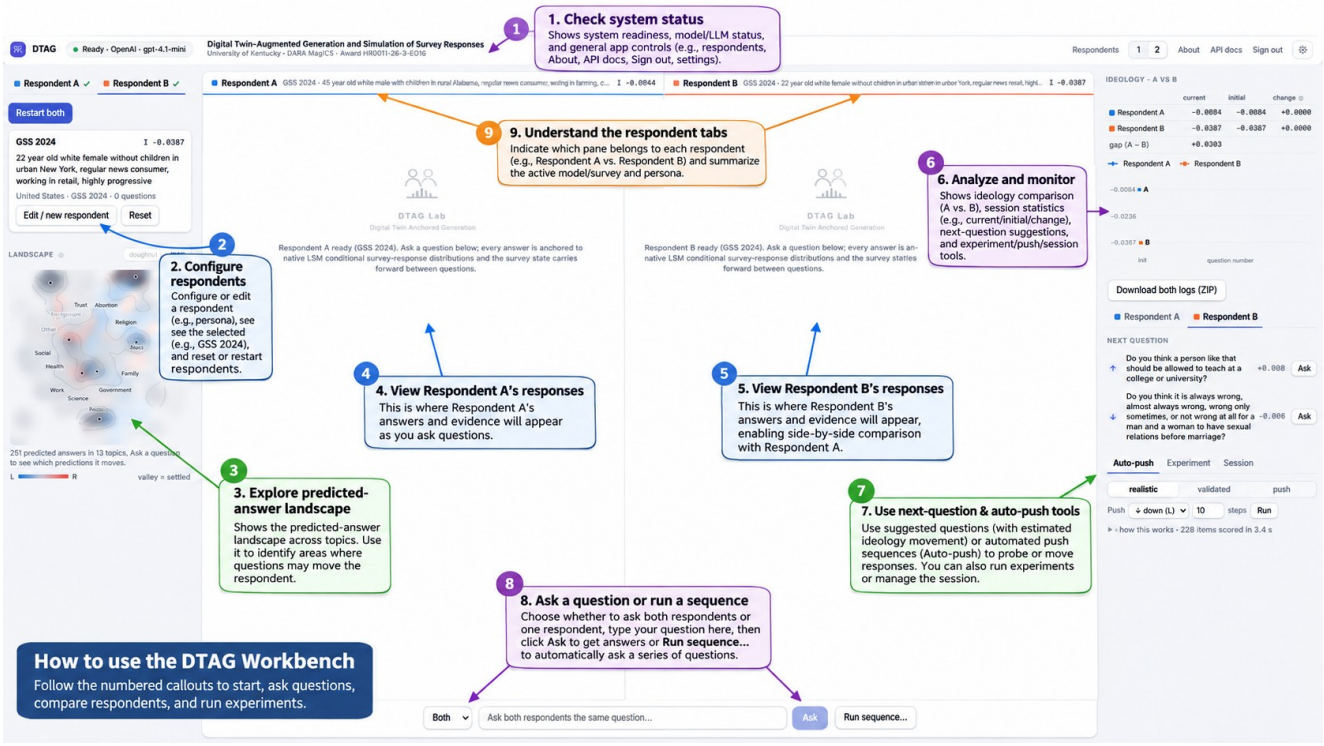


Figure 1. Annotated DTAG Workbench interface (two GSS 2024 respondents). The numbered callouts identify the system status, respondent configuration, predicted-answer landscape, parallel response panes, ideology comparison, next-question and auto-push tools, question composer and respondent tabs.

Source: DTAG repository, docs/dtag-workbench-annotated-guide.png (user-supplied original image).

2.3 Presentation and resizing

The workbench supports light/dark themes and drag-to-resize side columns. Double-click a divider or use “reset layout” to restore defaults. Below a narrower screen width, the columns stack and some next-question chips move above the composer. Figure export controls appear when hovering the relevant visualization.

3 Choosing a survey and constructing the respondent

The survey and its time/geographic coverage determine which native conditional distributions can be queried.

3.1 Who, Where, When

Who. Enter a concise demographic and social description (age, occupation, education, household, general attitudes as appropriate). The model interprets this into valid survey item/answer assignments; it does not turn every descriptive word into a model constraint.

Where — country. Enter a country when appropriate. A recognized country in the description may be pre-detected; an explicit field overrides it. Whether the country is hard-conditioned or contextual depends on the survey family.

When — year. Enter the relevant survey year. For GSS, the year selects a discrete wave; for WVS7 it can condition the pooled model’s year variable; for Afrobarometer it selects the nearest round.

Exact date. Use the date disclosure when routing Eurobarometer. An exact fieldwork date must resolve unambiguously; a calendar year alone does not identify a single Eurobarometer study.

3.2 Survey choice and conditioning

Survey family	Model/time selection	Geography and caveat
GSS	35 discrete US waves, 1972–2024; nearest requested wave.	US national sample; state/city/country are persona context, not a geographic hard constraint. GSS ideology is available.
WVS7	One pooled model for 2017–2023; survey year can condition A_YEAR.	Country proxy through model-supported coordinates; check resolved conditioning.
Afrobarometer	Separate rounds R1–R9, approximately 1999–2023.	Covered African countries can be hard-conditioned via categorical country field.
Eurobarometer	207 ZA studies; explicit ZA or unambiguous exact date.	Country values are model-dependent; ambiguous subnational splits remain contextual.

DTAG recommends an available model, preferring temporal fit, directness of geographic conditioning and recency. The default is a recommendation, not a guarantee of best scientific comparability. Researchers can override it with “Choose a specific native model…” and should document the resulting model identifier, time and conditioning mode.

3.3 Presets, custom profiles and model readiness

1. Use an existing preset or enter a new description. Examples in the documented interface include contrasting GSS 2024 personas.
2. Inspect the recommended survey family and model key. The interface may display download, verification, extraction, loading and loaded states.
3. Expand “Survey response behaviour” and choose response and fallback policies before starting.
4. Select “Start respondent”. Examine the initial survey state and, for GSS, the baseline index at step 0.
5. Use “Save as custom profile” for a reusable configuration, or “Advanced profile builder…” to choose additional survey/map settings. Custom profiles on Cloud Run are not guaranteed to persist across replacement of the service instance.

3.4 Initialization is a constraint set, not a person

Persona interpretation proposes assignments only from response labels supported by the chosen model. Incompatible assignments are dropped. Hard geographic/temporal assignments, where supported, take precedence. The initial state should be inspected for unexpected or missing conditioning, especially before controlled experiments. A near-zero starting ideology index is not evidence of a politically neutral human; it is a relative model geometry measurement.

4 Ordinary survey interaction and model evidence

Natural-language questions are mapped to the survey’s own items, answered from native conditional distributions, and recorded with provenance.

4.1 Asking a question

1. Enter a question in the center composer or select a directly mappable starter suggestion.
2. If two respondents are enabled, specify Both, A only or B only.
3. Submit with “Ask”. Read the generated answer and its mapping/status tag.
4. Expand “Model evidence” to inspect the native survey anchors, distributions and selected response labels.
5. Check whether the state updated and, for GSS, how the index changed; repeat with a follow-up question.

4.2 Mapping, distribution and anchoring

The normal question path filters the survey’s semantic map for candidate variables, uses language-model selection to identify relevant survey items, and asks the native LSM for each conditional response distribution P(A_v=a | s). The session then selects the native response by the configured mode: “draw” samples probabilistically, whereas “max” takes the modal response. The answer is phrased to reflect these anchors.

A single natural-language question may map to several survey variables. Consequently, one conversational turn may create multiple survey assignments and downstream prediction changes. A question that is absent from the source questionnaire should not be treated as if the LSM had measured it directly.

4.3 Mapping classifications and fallback policy

Status	Interpretation	Effect on state
DIRECT	The question was grounded in one or more direct survey-variable matches.	The resulting native anchors update the survey state.
SEMANTIC / ANSWER ONLY	Indirect bridge items supported a prose answer but were not accepted as state observations.	No survey-state mutation from the bridge.
SEMANTIC / STATE UPDATE	Indirect bridge items supported the answer and are inserted as observations.	State updates from the bridge answers.
NO MATCH	No sufficiently defensible item was selected under the active policy.	No valid item-based state update.

The documented web configuration defaults semantic fallback to “update_state”; “answer_only” is the more conservative choice when indirect semantic bridges should not influence subsequent predictions. Set “off” for strict direct-item analyses, accepting that more questions may return NO MATCH. For research, report the chosen fallback setting and audit each bridge.

4.4 Reading “Model evidence”

- Mapping: which native variables were selected; direct versus semantic fallback; selection rationale.
- Question provenance: native survey wording, semantic map and (where relevant) fallback map origin.
- Conditional distribution: all supported answer labels and their probabilities under the current survey state.
- Anchor: the selected label and its probability; response mode and possible stochastic draw.
- State updates: new or overwritten assignments; whether an old entry was evicted at the state-size cap.
- Conditioning: survey family, wave/year, geographic method and hard constraints (or context only).
- Timings: route/inference/generation time where displayed.

4.5 Interpreting changes within a session

If the same question is asked after a different history, the native conditional distribution may differ because the state differs. The visible evidence is essential for distinguishing a genuinely different model prediction from an unrelated change in phrasing or mapped survey items. With deterministic “max” anchors and unchanged state, repeated native inference should not be interpreted as stochastic variation; persona interpretation or mapping can still vary if rerun independently.

5 Decision landscape: all survey families

The landscape summarizes native predicted responses to askable survey items before and after every interaction.

5.1 Doughnut view

The outer ring contains a wedge for each unanswered item, grouped in topic arcs. Wedge length reflects prediction concentration (“settledness”), operationalized as 1 minus normalized response entropy. The inner ring displays already-anchored items as beads. After a new answer, wedges with changed distributions pulse and display a spike reflecting change magnitude; the interface reports the most affected topics and items.

For a GSS model with poles, wedge color can represent one-step ideology lean (blue toward L, red toward R, neutral gray). For other survey families, the landscape is available without ideology lean coloring. These colors describe computed conditional geometry, not a survey respondent’s personal affiliation.

5.2 Terrain/map view

The alternative map places related items near each other in a fixed two-dimensional layout informed by model dependencies and topic groupings. Terrain height denotes indecision (high ground means relatively uncertain predictions; settled predictions appear as valleys). Answered items are pins. Recent affected predictions are animated near the new pin.

5.3 Live update behavior

A model answer triggers recomputation of items that can depend on the changed assignments. Dense non-GSS models may calculate the landscape in the background, progressively updating the visualization while questions remain usable. An intermediate view may temporarily mix updated and previous values; interpret the completed view as the current state. The final displayed predictions are computed by the native model, not generated as psychological explanations.

5.4 Cross-survey “next question” assistance

Without GSS ideology, the interface can recommend “open” items (less settled conditional predictions) and “connected” items (items likely to shift other predictions). These are landscape-based information probes and should not be described as directional ideological pushes.

6 Comparing two digital respondents

Respondents A and B maintain separate models, states, seeds and histories, even when posed identical questions.

1. Switch the toolbar from “Respondents 1” to “2”. The second respondent uses an independent setup and server session.
2. Select the A or B setup tab. Configure the same survey wave for controlled persona contrasts, or different models for explicitly cross-survey comparisons.
3. Select “Start both respondents” or start each individually.
4. Use the composer’s Both / A only / B only selector to control exposure to an individual question or ordered sequence.
5. Compare each conversational answer using its Model evidence; for GSS, the right column overlays trajectories and shows the A − B index gap.
6. Export the paired record before reset. Switching back to one-respondent view hides B without terminating B’s current session.

The A index uses its own selected model’s pole construction; B uses its own. A − B is most readily interpretable when both use the same wave, pole definition, and comparable initialization. Across different GSS waves, normalized index values are wave-relative and differences are not automatically longitudinal evidence. Across different survey families, no common ideology-axis score is provided.

7 GSS-only ideology index: definition and interpretation

GSS provides a wave-specific model geometry and a numeric index relative to registered left and right survey-answer pole states.

7.1 Formal definition

Let s be the respondent’s current survey state, L and R the designated wave-specific left and right pole states, and d(·,·) the native LSM qdistance computed from differences in conditional survey-response distributions. DTAG reports:

$$I(s) = [\, d(L, s) - d(R, s) \,] / d(L, R)$$

Thus I(s) increases as the current state becomes relatively closer to R rather than L. A positive movement ΔI denotes movement toward the R pole; a negative movement denotes movement toward L. A near-zero value means approximately equal relative qdistance to the two designated poles—not absence of ideological structure or an objective party-independent “moderate” identity.

7.2 What qdistance measures

The workbench’s native qdistance compares model-predicted response distributions, using Jensen–Shannon-divergence-based discrepancies aggregated across survey items. The index therefore places a partial answer state within the survey model’s conditional-response geometry. It is not a score calculated by assigning a partisan valence to the language model’s prose or merely counting answers containing ideological terms.

7.3 Baseline, per-turn movement and trajectory

“Initial ideology” is calculated after persona/temporal/geographic initialization. After any state-updating answer, the index is recomputed and added to the trajectory. For a turn from state s to s′, $\Delta I = I(s') - I(s)$; net change over a sequence is I(s_final) − I(s_initial). The session chart, individual push tooltips and sequence summary expose different levels of this record.

7.4 Comparability and safeguards

- Use the same GSS wave, pole set and response policy for within-experiment comparisons when possible. Pole states are wave-specific, so different-wave indexes are not automatically equivalent scales.
- Record persona initialization and any assigned ideology-related answers. Baseline scores can reflect those assignments before a single interactive question is asked.
- Do not equate “right pole” with Republican recruitment or “left pole” with Democratic recruitment; affiliation, self-reported political views and inferred model geometry are distinct variables.
- Index changes are conditional simulation results; no human persuasion was measured by the workbench.
- With the same final set of assignments and no state eviction, the final index is a function of final state, not the order in which that set was entered. Ordering can still affect intermediate distributions, sampled answers, adaptive question selection and final composition.

8 GSS next-question optimizer and push modes

The GSS optimizer searches unanswered opinion variables for questions predicted to increase or decrease the current ideology index.

8.1 One-step scoring

For an eligible unanswered item v and a possible native answer a, DTAG can evaluate the counterfactual state $s \cup \{v=a\}$. The response-specific one-step movement is:

$$\Delta_v(a \mid s) = I(s \cup \{v = a\}) - I(s)$$

In sampled (“draw”) response mode, optimizer ranking uses the probability-weighted expected movement, $\mu_v(s) = \Sigma_a P(a \mid s, v)\, \Delta_v(a \mid s)$. In modal (“max”) mode, the predicted movement corresponds to the response with the largest conditional probability. The optimizer shows one recommended upward question and one downward question, refreshed as the state changes.

Eligibility excludes administrative, demographic and factual-behavior items from the default opinion-question search. Pole-defining items are excluded unless “Allow pole-defining items” is enabled. The text shown to the user is formulated as a neutral question for the targeted native survey item.

8.2 Inspecting a suggestion

Click the suggested text to open its details before using “Ask”. The disclosure lists the targeted variable and survey wording, whether it is a pole item, the predicted index change, the basis of the score, and all available native responses with their probabilities and response-specific Δ index values. Validated-mode disclosures also show which candidates passed or failed the mapping-back check.

8.3 The three modes are substantively different

Mode	Mapping path	How to interpret prediction
realistic (default)	The neutral wording is processed like any typed question through the normal semantic map. It may map to additional items or miss the proposed target.	Best approximation to normal question asking; predicted target-only Δ and realized total Δ need not agree.
validated	Top candidate questions undergo a mapping dry run; failures to map back are discarded. Remaining candidates are ranked for the complete checked mapping.	Tests that the worded question reaches its intended item. Under max, mapped deterministic Δ can match the predicted value.
push (direct)	The optimizer-selected native item is answered directly; natural-language mapping is bypassed.	An idealized model-directed perturbation, not a realistic prompt–mapping test. Under max and no eviction, the chosen item's Δ is exact.

8.4 Predicted versus realized movement

The chart records the realized state and therefore the realized index. With draw, the selected answer can be any native response, so Δ may vary from the expectation even in direct mode. With realistic mode, mapping may hit several items or miss the target entirely. When the survey state reaches its retention limit, eviction of an old assignment can alter the net movement. The chat tags targeted turns (PUSH up/down) and may flag a missed item; inspect the tooltip and evidence for actual mapped variables, predicted Δ and realized Δ.

8.5 Recommended protocol for scientifically interpretable push claims

1. Choose a fixed GSS wave and characterize the initial persona/state.
2. Predefine the response mode: max isolates deterministic predicted movement; draw represents conditional sampling variability.
3. Use realistic when evaluating normal prompt-to-survey routing, validated when requiring demonstrated mapping fidelity, and direct push only for an upper-bound/idealized model-coordinate analysis.
4. Retain "Allow pole-defining items" unchecked for a conservative natural-opinion-item search, unless inclusion is part of the experimental design.
5. Record mapping hit rates, all mapped items, predicted and realized movements, and any early termination or eviction.

9 Automatic GSS push

Auto-push iterates the same directional optimizer, recalculating the best next item after each state change.

9.1 Running an automatic push

1. Initialize a GSS respondent and confirm "Ideology tracking" is enabled.
2. In the right column, open "Auto-push". Choose realistic, validated or push mode.
3. Choose ↑ up (R) or ↓ down (L). The UI accepts 1–100 requested steps; its initial suggested count is five.
4. Optionally enable pole-defining items only if the design requires them.
5. Select "Run". Monitor the progress card and live trajectory. Cancel if needed; cancellation takes effect after the current in-flight question.
6. Inspect the resulting trajectory, actual questions asked, mapping outcomes, final index and export.

9.2 What the algorithm does and does not hold fixed

At each step the tool scores presently eligible unanswered items using the respondent's new state, chooses the item expected to move in the requested direction, asks/anchors it using the selected mode, and repeats. Consequently, the sequence of native items is adaptive rather than a fixed list. The process may stop before N when no further question yields the requested movement. Results depend on initial state, response mode, language-based mapping behavior, available opinion items and random seed.

9.3 Design caution

An increase in the internal GSS index after directional question selection is not evidence that the software has persuaded someone. An optimized adaptive information-acquisition sequence preferentially conditions on survey answers that shift a fixed learned conditional geometry. For a fair assessment, compare against non-optimized questions, opposite-direction pushes and independently sampled reruns, holding survey wave and baseline construction constant.

10 Ordered question sequences

Sequences allow controlled lists of natural-language questions on any survey, independent of GSS auto-push.

10.1 Preparing a question list

Select “Run sequence…” next to the composer. Paste questions one per line or upload a text/CSV file. For CSV, the column name “question” is recognized; other columns can be present. Blank lines, lines beginning with # and a header line “question” are ignored. The documented limit is 500 questions.

10.2 Running and monitoring

1. Choose whether to reset the respondent first; otherwise the sequence starts from the current state.
2. Start the ordered sequence. Each question traverses the ordinary mapping, native anchoring and state-update path.
3. Monitor completed/remaining questions and answers as they appear. During execution, interactive asking and reset are temporarily unavailable.
4. Cancel when necessary; completion/cancellation is reported after the currently executing question.
5. For GSS, review starting/ending ideology, net change, range and the questions with the largest index movements; for any survey, inspect the complete per-question evidence and export.

10.3 Fixed versus adaptive protocols

A fixed list can be used to study a reproducible input order, although realized anchors can still vary with draw sampling and persona initialization. An auto-push instead adapts which native survey item to ask after every update. These are different study designs. In particular, interpreting order effects requires checking whether the final assigned item–answer state differs, rather than inferring an intrinsic noncommutative ideology metric.

11 Experiment recipes and repeated runs

The Experiment tab converts a live interaction history into a replayable research protocol and produces run-level records.

11.1 What is recorded

The right-column Experiment tab automatically assembles a recipe from typed questions, optimizer-suggested questions, manual pushes, auto-pushes and ordered sequences. The action disclosure shows the recorded list. A reset starts a new recipe. Pushes and suggestions are recorded as action intent: on replay the optimizer can select different items at the state reached by that replicate.

11.2 Rerun ×N workflow

1. Construct the intended protocol in a live session. Expand the recorded actions and confirm their order and type.
2. Enter an optional experiment name and the number of independent runs (documented UI range 1–200; default 20).
3. Click “Rerun ×N”. Each run starts a fresh respondent with the same profile and a run-specific seed.
4. Observe queue/running status, completed runs, live progress and errors. You may cancel a running batch.

5. Inspect the summary visualization and “Questions asked per action”; compare mapped targets/hit rates, not just the net ideology shift.
6. Download both the recipe JSON and the final experiment results ZIP for independent analysis and archiving.

11.3 What varies across replicates

The recorded action policy is replayed; the exact anchored answer and therefore the state trajectory may vary. The new persona initialization can also vary because it is independently interpreted, and in draw mode each run has its own seed (base seed plus run index). For an optimizer-based action, the actual suggested question is re-selected at that run’s current state. This gives a distribution over model-based trajectories, not a direct estimate of the sampling uncertainty of a real-world intervention.

11.4 Summary statistics

Result	Meaning / interpretation
Per-run trajectory	Index after successive recorded actions, shown as faint run-level lines (GSS).
Mean index and 95% bootstrap CI	Across-run mean conditional simulation trajectory and bootstrap uncertainty band, computed over generated replicate runs.
Net index change and CI	Mean of final minus initial simulated index; variation includes draw/initialization and adaptive replay effects.
SD, n	Variation and number of experiment replicates included.
Push hit rate	Share of targeted push questions for which normal mapping reached the intended item, when applicable.
Questions asked per action	Frequency of the actual chosen question text, distinct choices and skipped actions across runs.

For non-GSS models, repeated runs and answer/question records remain available, but the Experiment panel cannot display an ideology trajectory because no index exists. Its text reports run completion and points to the downloaded records for further analysis.

11.5 Critical experiment comparisons

- Baseline-control comparison: report the initialization strategy and baseline index distribution. Fresh persona interpretations can introduce baseline differences unless explicitly controlled in the design.
- Mode comparison: realistic versus validated versus direct answers different scientific questions; compare all three only with the mode distinction made explicit.
- Direction comparison: up, down and neutral/fixed-list protocols provide a check on algorithmic directional selection; use matched survey wave and persona design.
- Question-history and order comparison: preserve exact questions and answers as well as aggregate index. Reordering with identical final assignments cannot by itself change a state-defined terminal index.
- Report individual-run outcomes and 95% CIs; do not treat every turn inside one replicate as an independent participant.

12 Exports, figures and reproducibility

For the hosted service, downloading the outputs is the critical persistence mechanism.

12.1 Session → Export

In the right-column Session tab, open “Export ▾”. The documented formats are Full log (ZIP), JSON and CSV. The full ZIP can include a human-readable report, question-level JSONL/CSV, initial/final survey states, state changes, persona initialization, settings, ideology trajectory and pole information (where applicable), sequence data, session summary, and a manifest with version/commit and model/checksum information.

12.2 Experiment → Download results (ZIP)

The experiment ZIP contains a recipe and manifest, a report, per-run traces and detailed states, plus aggregate summary files. The server-side folder name shown in the panel is not a substitute for downloading it; Cloud Run filesystem contents may be lost on instance replacement.

Typical artifact in ZIP	Use
recipe.json	Reconstruct the replayed action policy, not necessarily fixed questions for adaptive actions.
manifest.json	Version/commit, model, run and seed metadata for provenance.
runs/run_NN/trace.jsonl	Question/action-level native anchors, mapped items, index before/after and push outcomes.
runs/run_NN/states.csv	Full accumulated survey state after individual turns.
runs/run_NN/state_changes.csv	Old-to-new answer assignment transitions.
summary/steps.csv, checkpoints.csv, stats.csv/json	Across-run step, checkpoint and statistical summaries.
summary/questions.csv, final_answers.csv	Actual asked questions and terminal model answers across runs.

12.3 Figure exports

Hover the ideology chart, doughnut, map or experiment plot and use the figure-save control. Available formats are SVG (vector), PNG (4× resolution) and PDF. Terrain visualization in PDF may contain a high-resolution raster because the map is drawn on a canvas. Store figures with the corresponding experiment ZIP, not alone.

12.4 Minimal reproducibility record

- Date/time, application “About” version and Git commit, model family/key and wave/ZA/round, map identity, model release and checksum if available.
- Full persona text; explicit geography/time; initial assignments; pole set (GSS); state_keep and fallback policy.
- Response modes, seeds, question wording/sequence, automatic mode and direction, whether pole items were eligible.
- Full experiment recipe, actual mapped variables, answer anchors and per-run traces; number of runs and CI definition.
- Any model loading errors, no-match turns, state evictions, interrupted actions and unsuccessful mapping-back checks.

13 Browser settings reference

Settings are part of a scientific protocol; record them alongside results rather than relying on a screenshot of the prose.

Setting (visible under behavior / advanced)	Documented default	Scientific effect
Response mode (anchor)	draw	Samples from the native conditional response distribution. “max” chooses its most likely answer.
Semantic fallback	update_state	off / answer_only / update_state control indirect bridge handling and whether state is changed.
Semantic response mode	max	Mode for native anchors selected during semantic fallback.
Seed	Profile or base 1000	Controls stochastic response draws, together with question index / replicate index.
Ideology tracking	on for GSS	Enables wave-specific index and directional GSS controls; unavailable for other surveys.
Max direct variables (k)	50	Upper bound on mapped native items from one question.
Lexical prefilter / min map score	200 / 1.0	Direct mapping candidate screening.
Max persona assignments / persona prefilter	50 / 500	Number of valid initial assignments and input candidate pool.
State keep	500	Max retained answer assignments; oldest

Setting (visible under behavior / advanced)	Documented default	Scientific effect
		may be evicted.
Semantic k / prefilter / min confidence	6 / 80 / 0.35	How many indirect bridge items may be considered and accepted.

The default language model is server-configured and displayed in the advanced behavior panel; it is not set by a user-supplied executable command or arbitrary endpoint. These defaults are drawn from the checked-in browser-workbench reference and should be confirmed against the “About” version and current behavior settings of the deployed service.

14 Worked examples: GSS and Afrobarometer

These two complete browser workflows show how to operate DTAG from respondent definition through independent records and export. Their personas, seeds, and question text are example inputs—not simulated findings. Record the actual native variables, probabilities, answer anchors, and ideology values shown by the running application.

Example 1 — GSS 2024: ordinary questioning, ideology push and a replicated experiment

Research objective. Construct one survey-grounded respondent from a defined GSS 2024 profile, observe ordinary mapped responses, examine the wave-specific ideological geometry, conduct a realistic directional push, and replay the recorded interaction across independent model-generated runs. This is a tutorial protocol, not a claim that actual humans change ideology.

A. Setup and a reproducible baseline

1. Open the password-protected Workbench and record About → version/commit. Select Respondents → 1. If a previous respondent is running, export its session before starting a replacement.
2. Choose preset gss2024_cm (the documented GSS 2024 CM example). Inspect the populated Who description rather than assuming its demographics from the preset name. Set or confirm Where = United States and When = 2024; choose native model gss/gss_2024.
3. Open Survey response behaviour. For this worked run choose Response mode = draw; select a seed (for example 1047) and record it, and keep Semantic fallback = update_state unless the protocol requires strictly direct matches. Confirm that Ideology tracking is enabled and record the polar set shown.
4. Review the model readiness indicator. Wait for download/verification/loading to finish if needed. Click Start respondent. In the Session tab, inspect the initial state and conditioning details; record the baseline ideology value I_0 displayed on the right. Do not substitute a presumed “conservative” value for the calculated index.

Expected display. The GSS-specific ideology chart starts at the initialized state, a decision landscape appears in the left rail, and two up/down next-question suggestions are computed. GSS is a US national survey: entering Alabama or another US locality as persona text does not hard-condition a state-level geographic survey variable.

B. Ask an ordinary question and audit its native evidence

5. Use the center composer and enter: “What are your thoughts about immigration?” This wording is an example of a natural-language prompt, not a guaranteed one-to-one survey item. Click Ask. Alternatively, click a starter question that explicitly identifies directly mapped native variables.
6. Expand Model evidence under the generated answer. Record the mapping class (DIRECT, SEMANTIC / ANSWER ONLY, SEMANTIC / STATE UPDATE, or NO MATCH); native variable code(s); original survey wording; possible answers and their conditional probabilities; selected response anchor; mapping rationale; and state changes.
7. Check the question mapping carefully. A broad immigration question may map to more than one native variable; if it returns NO MATCH, do not treat the prose as a measured immigration opinion. Use a more specific, suggested question and try again. When Semantic fallback is enabled, note whether the bridge changed state.

8. Read the updated ideology value I_1 and calculate $\Delta I = I_1 - I_0$ from the recorded chart values. Inspect the landscape: the doughnut and terrain views show changes in conditional predictions for unanswered items. Record the changed topics and top movers; this does not imply the respondent was persuaded.

Field to capture	Example worksheet entry — fill from the running app
Model / wave	gss/gss_2024; model release/version as displayed
Persona and initial assignments	Copied from Session export / initialization record
Question and mapping class	Immigration prompt; actual mapping class from Model evidence
Mapped item(s) and distribution	Variable codes, survey wording and complete P(answer | current state)
Selected answer and state update	Native answer anchor(s), probabilities, and changed state entries
Ideology checkpoint	I_0 = [observed]; I_1 = [observed]; ΔI = [calculated]
Landscape movement	Top affected survey variables/topics from the completed refresh

C. Inspect one step of the ideology optimizer

9. In the GSS right rail, open the suggested ↑ up (toward R pole) question and expand its detail row. Keep “Allow pole-defining items” unchecked for an ordinary opinion-item protocol. Read the target native item and neutral question text, the predicted ΔI, and the list of every possible answer with probability and per-answer ΔI.
10. With draw, interpret the displayed predicted move as a probability-weighted expected value: $\mu_v = \Sigma_a P(a \mid s,v) [I(s \cup \{v=a\}) - I(s)]$. The realized change depends on the sampled anchor. With max, the modal answer is taken and its calculated item-only move is the relevant prediction.
11. Select the realistic mode (the default) and click Ask beside the upward suggestion. This sends the generated question through the ordinary mapping step. Expand the resulting chat entry and compare the optimizer target variable with the actually mapped variable(s). Record whether the targeted item was hit and whether other items were also updated.
12. Record predicted and realized ΔI and any “missed item” indication. A discrepancy is not necessarily a model error: realistic mapping may reach additional items, skip the intended item, or use a sampled rather than modal answer.

Method comparison. For a controlled comparison, start fresh respondents from the same recorded profile and settings and repeat an analogous optimizer choice in validated and direct push modes. Validated checks question-to-item mappings before selection; direct push skips the mapping and forces exactly the target survey item. These modes answer different methodological questions and should never be pooled as equivalent treatment arms.

D. Run an adaptive push and reproduce it

13. In Auto-push, retain realistic mode, choose ↑ up (R) and select 5 steps for a quick demonstration (or a prespecified N for a formal study). Click Run. DTAG recomputes the suggested item after each state update; thus the sequence is adaptive rather than a fixed list.
14. Monitor the progress card, ask/anchor entries, and ideology chart. The process may finish before N when no remaining item has a beneficial directional move. Do not start an overlapping manual question or reset during the sequence. Cancellation takes effect after the current question.
15. Open Experiment. Expand the recorded actions and verify that both ordinary questions and directional pushes are represented, including intended direction and push mode. If the recipe includes exploratory steps, start a clean respondent and construct the intended protocol rather than replaying accidental actions.
16. Give the experiment a descriptive name such as “gss2024_cm_realistic_up5” and set Runs = 20 for a tutorial demonstration (the documented browser input permits 1–200). Click Rerun ×20. Each rerun starts a fresh respondent from the profile; seeded sampled responses and possible persona interpretations can vary, and optimizer questions are reselected for each run’s current state.
17. After the job completes, inspect the mean ideology-by-action trajectory, faint per-run trajectories, 95% bootstrap confidence band, net index change and its interval, push hit rate, and the distinct questions actually asked per action. The band describes variability across generated runs under this replay design; it is not a confidence interval for a causal effect in people.

18. Save the Experiment recipe (JSON), Download results (ZIP), and separately Session → Export → Full log (ZIP). Also save the ideology plot (SVG, PNG or PDF) and note the About version, initial state, selected mode, seed and model/pole metadata alongside the exported files.

E. Expected artifacts and interpretation

Where to look	What a successful walkthrough produces
Chat and Model evidence	Question text; mapping path; native conditional distributions and selected anchors
Ideology chart and push cards	I_0 and subsequent values; candidate predicted ΔI; realized ΔI; mode; mapping hit/miss
Experiment panel	Recorded recipe; number of runs; bootstrap band; net change; per-action question diversity
Session full log ZIP	Persona initialization, model/settings, answered items, state changes and ideology trajectory
Experiment results ZIP	Per-run question traces, state tables, summary/statistics files and reproducibility manifest

Scientific claim supported by this example. Conditional survey-model inference and adaptive information-selection behavior can be documented and reproduced for a defined GSS state. It does not establish persuasion or observable ideological movement in a human subject. A real participant study has a different protocol, sampling frame and inferential target.

Example 2 — Afrobarometer Round 3: Kenya 2005 survey-grounded inquiry

Research objective. Demonstrate a non-GSS survey workflow with explicit country conditioning, native survey-response evidence, landscape-based next-question selection, a fixed multi-question sequence, and a replayable experiment. This example has no GSS-style ideology score, directional pole optimizer or automatic ideological push.

A. Initialize a Kenya respondent and check geographic conditioning

1. At the Workbench, select Respondents → 1 (export the previous session first). Leave Preset as (none) or use a matching custom profile. Enter Who = “35-year-old primary-school teacher in Nairobi, Kenya, regularly listens to the radio”; Where = Kenya; When = 2005.
2. Review the recommended survey options and select Afrobarometer Round 3 (documented fieldwork period approximately 2005–2006). Confirm the exact model identifier, expected as afrobarometer/r3, in the Survey model area. An explicit different model selection overrides the recommendation.
3. Inspect the resolved country/time context. For this documented round, country is conditioned through a native categorical country variable (for example COUNTRY_ALPHA = Kenya, if this is the resolved field shown). Confirm the actual coding in the Session view; do not confuse the person’s Nairobi text with a city-specific native condition.
4. Select Response mode = draw with a recorded seed (for example 2047) and Semantic fallback = update_state for ordinary exploration, or off for a strict direct-map audit. Start respondent after the model is loaded. In the Session panel, capture the initialized survey state and the model/map identifier.

Expected display. Survey-grounded chat, Model evidence, the decision landscape, suggested next questions, sequences, experiments, and exports are available. GSS ideology trajectory and ↑/↓ directional Auto-push controls are not available for an Afrobarometer respondent.

B. Question answering and mapping audit

5. Ask: “How much do you trust the national parliament?” or select the closest suggested survey question. Inspect Model evidence for the actual native item code, wording, possible responses, probabilities, selected anchor, and whether the answer entered the state. The example wording is not a promise that a particular variable is present.
6. Ask: “How satisfied are you with the way democracy works in Kenya?” Again record the mapping class and response anchor. A direct match means the survey model was asked about mapped items; a semantic bridge must be identified as such. If NO MATCH appears, use an exact suggested question instead of assigning a substantive answer.

7. Switch the landscape from doughnut to map. In the doughnut, wedge length reflects predicted answer certainty, grouped by topic; in the terrain view, the modeled geography of survey-variable dependencies is shown as valleys/high ground, not a geographic map of Kenya. Compare which unanswered items' predicted distributions moved after each anchor.

C. Use non-GSS suggestions and a controlled sequence

8. Review the "open" and "connected" next-question suggestions when available. "Open" targets relatively uncertain predicted responses; "connected" targets questions that would affect many other conditional predictions. Neither category is a left/right ideology recommendation.
9. Click Run sequence… and paste the following three questions into its question-list text area. Blank lines and comments are ignored; the documented UI also accepts one-question-per-line text files or a CSV with a question column. For a clean sequence comparison, choose the option to reset before running.

- Question: How much do you trust the national parliament?
- Question: How satisfied are you with the way democracy works in Kenya?
- Question: How often do you discuss politics with friends or family?

10. Start the sequence and monitor the progress card. The three natural-language prompts pass through the normal mapping/fallback route in the stated order. Each accepted state update conditions later predictions. The sequence can be cancelled after the active question, but cannot be edited halfway through without creating a new run.
11. After the sequence, open individual Model evidence entries and verify which native variables were actually asked. If any question is NO MATCH or a weak semantic bridge, do not treat this as a validated three-item Afrobarometer instrument. For direct-item experimental work, replace that question with one from DTAG's map-grounded suggestions and rerun from a reset state.

D. Repeat, export and analyze without an ideology index

12. Open Experiment and expand the recipe. Check that it contains the requested three-question sequence (plus any exploratory actions taken since the most recent respondent reset). Start a fresh respondent if the recipe contains unwanted exploratory turns.
13. Select an experiment name such as "afro_r3_kenya_sequence3", choose 20 runs for the tutorial and click Rerun ×20. Each run reconstructs the profile and replays the recorded question intent with its own seed. Because this family has no registered ideology poles, the results panel reports completed runs and question counts rather than a GSS index trajectory or directional net change.
14. Expand Questions asked per action, examine whether the same question mapped to similar survey items across runs, and download the results ZIP. Export the current respondent session separately (full log ZIP, JSON or CSV). Record model/round, geographic conditioning, semantic fallback, seed, all mappings, native answers, and sequence order.

Available in this non-GSS example	Not available or not equivalent
Native response distributions, anchored answers, mapped item identifiers	No GSS ideology I(s) computed
Doughnut / terrain decision landscape and affected-item summaries	No red/blue left/right pole interpretation
Open/connected suggested questions	Not the GSS optimizer's ↑/↓ recommendations
Question sequence, recipe, rerun count and per-run trace ZIP	No ideological net-change bootstrap plot
Direct and fallback mapping audit; country conditioning metadata	Not a causal political opinion-change experiment

Scientific claim supported by this example. For the chosen Afrobarometer round and Kenya-conditioned native state, DTAG provides conditional distributions and transparent response evidence under sequential questioning. Comparing conditional predictions across turns is a model-based dependency analysis; it is not longitudinal measurement of the same population or a causal intervention.

15 Troubleshooting and operational cautions

Symptom	Likely explanation / action
Sign-in expired / unauthorized	Return to the login screen. The shared-password cookie can

Symptom	Likely explanation / action
	expire; server-side session survival is not guaranteed after service replacement.
Model pending / slow first use	The native archive may be downloading, checksum-verifying, extracting or loading. Check readiness in the survey model panel.
Question gives NO MATCH	No direct item and no accepted semantic bridge under the active policy. Try a survey-measurable wording; do not force an unwarranted response.
Unexpected state change	Inspect direct mapping, bridge classification, multiple mapped variables and state eviction in Model evidence.
Expected and realized push differ	Check response mode draw/max, mapping mode, target hit, additional variables and retention-limit eviction.
No GSS index or push tools	Verify a GSS model with ideology enabled and registered wave-specific poles; those functions are not general to other families.
Eurobarometer year not selecting study	Provide a fieldwork date or choose a specific ZA; a calendar year is ambiguous.
Landscape temporarily stale	Dense models update predictions progressively in background; let updates complete before comparing final landscapes.
Reset or app restart lost work	Respondent state and server files are not durably guaranteed on hosted Cloud Run. Use exports before resetting or ending.

15.1 Data handling and access limits

The documented service uses a shared password rather than individual accounts and has no guaranteed persistent participant/session database for the Workbench. Avoid entering identifiable or sensitive personal data as persona text. Treat exported sessions and experiments as research artifacts and apply institutional storage and access controls outside the web application.

15.2 When the document and the UI differ

Treat the live “About” version/commit and visible capabilities as decisive. This manual was built from the connected DTAG repository’s browser-facing documentation, not from a completed authenticated interactive audit of the currently served build. Record inconsistencies and revise the manual for a particular deployed release before external distribution.

16 Interpretation checklist for reports and publications

A technical report using DTAG should state what the digital respondent represents, the survey/model wave, the current state, what was directly mapped versus semantically bridged, the response mode, and the precise inference being made. For GSS it should also define the pole states and qdistance-based index and distinguish selected predicted changes from actual realized index movement in the simulated session.

A replicated GSS push analysis should identify whether the question path was realistic, validated or direct, whether answers were sampled or modal, how often mapping hit its intended item, and whether the interval is a bootstrap interval over model-generated reruns. A causal claim about human attitudes would require a separate human experimental design and is not a result of the Workbench alone.

Suggested concise result language: “Within the specified GSS-wave native LSM, adaptive selection and conditioning on survey-response anchors shifted the model-derived ideology index by the reported amount. The statistic summarizes conditional simulation trajectories, not observed persuasion of human participants.”

A Appendix: glossary

Term	Meaning
Anchor	Native survey response selected for a matched variable from the conditional distribution.
Askable item	Native item available for questioning and not already excluded/answered for the current operation.
Conditional distribution	Model estimate of response-label probabilities given current survey assignments.

Term	Meaning
Direct mapping	Question mapped to surveyed variable(s) through the normal semantic-question selection.
Ideology index	GSS-only pole-distance contrast of the respondent's partial state.
Mapping-back check	Validated-mode test that generated wording maps to the intended native variable.
Native LSM	Survey-wave/pooled native model producing item-level conditional distributions.
Pole-defining item	Native response item used in construction of a GSS left/right pole state.
Push hit	A targeted question whose normal mapping reaches the item it intended.
Recipe	Recorded interaction actions that can be replayed on fresh respondents.
Semantic bridge	Indirect survey-item connection used when no suitable direct match is found.
Survey state	Current set of survey item/answer assignments that conditions predictions.

B Appendix: reference sources and release applicability

Primary checked-in sources for this user manual (connected GitHub repository zeroknowledgediscovery/DTAG, branch main, retrieved 9 November 2026):

- User-facing platform description and scientific reference: https://github.com/zeroknowledgediscovery/DTAG/blob/main/README.md
- Browser Workbench flows, GSS optimizer, experiments, landscape, exports and dual respondents: https://github.com/zeroknowledgediscovery/DTAG/blob/main/webapp/README.md
- First-use browser workflow: https://github.com/zeroknowledgediscovery/DTAG/blob/main/QUICKSTART.md
- User-visible GSS push mode descriptions and limits: webapp/frontend/src/components/IdeologyPush.tsx
- Experiment replay UI and statistics: webapp/frontend/src/components/Experiment.tsx
- Survey setup and behavior controls: webapp/frontend/src/components/RespondentSetup.tsx and Controls.tsx
- Hosted session and access caveats: https://github.com/zeroknowledgediscovery/DTAG/blob/main/CLOUD_RUN.md

Source observation: the repository file references consulted resolve to commit 8afdbb596dc88ccb60a5ca630e05fba111de2b9a. A separate earlier deployment report referenced an image tag 19fd614; comparison of the two repository revisions found no substantive changes to the documented browser UI components. Because direct access to the protected browser deployment was unavailable for validation, this draft does not certify that the running service exactly matches these checked-in sources.

Pre-release review checklist

- Log into the hosted application and record About/version/commit.
- Confirm the deployed interface includes each named control and the documented settings defaults.
- Test one GSS respondent, one non-GSS respondent, one two-respondent session, one sequence and one rerun.
- Check the actual ZIP structure and a figure export against the artifact descriptions.
- Replace this verification note with the observed deployment revision and release date; only then issue a production manual.